

■ Homework #4: 2025학년도 1학기

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■ 고속 푸리에 변환을 임의의 관심 이미지에 적용한 예를 보이시오.

```
import numpy.fft as fp
from skimage.io import imread
from skimage.color import rgb2gray, rgba2rgb
import matplotlib.pyplot as plt
import numpy as np

def plot_image(image, title=''):
    plt.imshow(image)
    plt.axis('off')
    plt.title(title, size=20)

img2 = imread('/content/drive/MyDrive/task/Guangan Beach.jpeg')
if img2.shape[-1] == 4: # Check if the image has an alpha channel (RGBA)
    img2 = rgba2rgb(img2) # Convert RGBA to RGB
img2 = rgb2gray(img2)

freq2 = fp.fft2(img2)
iminv2 = fp.ifft2(freq2).real

shiftfreq2 = fp.fftshift(freq2)
proj2 = 20*np.log10(0.01 + np.abs(shiftfreq2))

plt.gray()
plt.figure(figsize=(12,10))

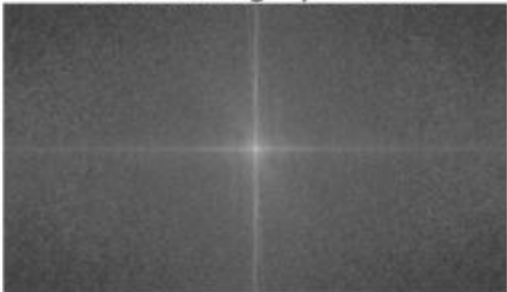
plt.subplot(2,2,1).plot_image(img2, 'Original Image')
plt.subplot(2,2,2).plot_image(proj2, 'Fourier/Mag Spectrum')
plt.subplot(2,2,3).plot_image(np.angle(shiftfreq2), 'FFT Phase')
plt.subplot(2,2,4).plot_image(np.clip(iminv2,0,255), 'Reconstructed Image')

plt.show()
Figure size 640x480 with 0 Axes>
```

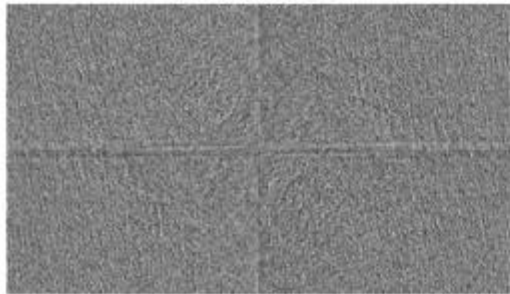
Original Image



Fourier/Mag Spectrum



FFT Phase



Reconstructed Image



■ 문제 발생시 자신 만의 해결 방법 서술

실습으로 인해 고속 푸리에 변환이 영상처리에서 어떻게 사용되는지 알게 되었다.

[용지 부족시 추가하여 작성]

[캡처 예) : 코드와 실행화면이 같이 나오도록 캡처

```
import numpy.fft as fp

img2 = imread('house2.png')
if img2.shape[-1] == 4: # Check if the image has an alpha channel (RGBA)
    img2 = rgba2rgb(img2) # Convert RGBA to RGB
img2 = rgb2gray(img2)

freq2 = fp.fft2(img2)
iminv2 = fp.ifft2(freq2).real

shiftfreq2 = fp.fftshift(freq2)
proj2 = 20*np.log10(0.01 + np.abs(shiftfreq2))

plt.gray()
plt.figure(figsize=(12,10))

plt.subplot(2,2,1), plot_image(img2, 'Original Image')
plt.subplot(2,2,2), plot_image(proj2, 'Fourier/Mag Spectrum')
plt.subplot(2,2,3), plot_image(np.angle(shiftfreq2), 'FFT Phase')
plt.subplot(2,2,4), plot_image(np.clip(iminv2,0,255), 'Reconstructed Image')

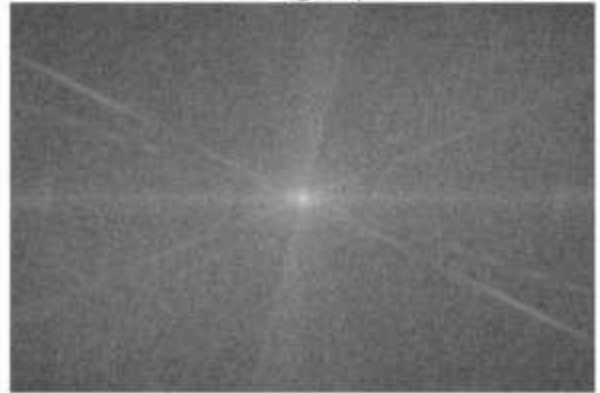
plt.show()
```

<Figure size 640x480 with 0 Axes>

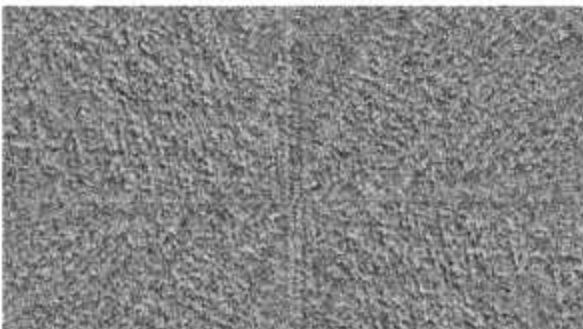
Original Image



Fourier/Mag Spectrum



FFT Phase



Reconstructed Image

